Amna Munawar

Data Engineer | Data Platform | DataOps

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Summary

Senior Data Engineer with end-to-end ownership of AWS data lakehouse platforms, architected real-time streaming and batch pipelines (Kafka, Spark, Trino, dbt) processing 65M+ records, delivering 30% cost reduction, 45% faster pipelines, and sub-second query SLAs. Deploys AI-powered data infrastructure with platform-level ownership of data quality and reliability.

Experience

Bazaar Technologies **July 2025 - Present**

Senior Data Engineer II Karachi, Pakistan

- Architected end-to-end **data lakehouse** platform (Spark, Kafka, Airflow, Trino, dbt, S3, **Iceberg**) processing 10M+ records daily, reducing time-to-insight by **70%** across 10 business teams.
- Engineered **real-time streaming** pipelines using TimescaleDB hypertables and continuous aggregates on Kubernetes (CNPG), reducing data latency SLA from 20 minutes to under 1 minute and driving a **40%** sales increase.
- Deployed an **AI agent**-powered **NL-to-SQL** platform using **RAG** (OpenCode + Open WebUI + Trino on Kubernetes) over a medallion architecture, enabling **self-service analytics** and cutting analyst workload by **60%**.
- Built Python-based **hot/cold tiering** pipelines with automated lifecycle management for 65M+ MongoDB documents to Apache Parquet, cutting storage costs by **30%**.
- Orchestrated zero-downtime data migration using **CNPG**, **Helm**, and AWS DMS over VPC PrivateLink, maintaining **99.9% uptime SLA** during cutover.
- Implemented Hive Metastore as a centralized **schema registry** with **data contracts** across Trino and dbt, cutting schema inconsistencies by 70% across **DataOps** pipelines.
- Deployed observability stack (Percona + Grafana) with automated alerting for PostgreSQL and MySQL RDS, reducing database incidents by **75%** and achieving **2x** faster incident resolution.
- Collaborated cross-functionally with AI and DevOps teams to provision AWS resources, build CI/CD pipelines, and deliver clean datasets for LLM-powered chatbot serving.
- Mentored interns and junior engineers through structured onboarding roadmaps and pair programming, accelerating ramp-up on data platform workflows.

Senior Data Engineer I **January 2024 - June 2025**

- Optimized Trino cluster performance via **FinOps** practices (KEDA autoscaling, Kubernetes resource tuning, query routing), cutting infrastructure costs by **25%** and boosting throughput by **40%**.
- Enforced **data governance**, **data lineage**, and GDPR compliance across **ELT** pipelines via Apache Ranger (column-level masking, RBAC), eliminating **100%** of manual user provisioning.
- Improved Airflow **pipeline reliability** by refactoring resource-heavy DAGs with modular Python design and batch processing optimization, reducing SLA breaches by 80% and raising **pipeline SLO** to **99%**.

Software Engineer II **January 2023 - December 2023**

- Migrated legacy Spark workloads to a Trino and dbt medallion lakehouse with optimized ELT transformations and Python orchestration, cutting pipeline runtime by **45%** and compute costs (FinOps) by **35%**.
- Optimized AWS RDS (MySQL) performance for 50M+ records by implementing table partitioning and automated Python/SQL archiving strategies, delivering **3x** query speedup and **40%** storage reduction.
- Built a data quality and discovery platform using OpenMetadata (cataloging & lineage), dbt (schema & data validation tests), and Metabase (monitoring dashboards & automated alerting) reducing data quality incidents by **95%** .
- Automated Infrastructure as Code (IaC) deployments with Terraform and ArgoCD (GitOps), eliminating manual provisioning and accelerating release cycles by **70%**.

Software Engineer I **July 2022 - Dec 2022**

- Replaced full RDS snapshot backups with a DuckDB-powered partition-level Parquet pipeline on S3 via Airflow, cutting storage costs by **60%** and reducing restoration time from hours to minutes validated during a critical production incident recovered in under 2 hours.
- Built microservices on Kotlin, Java and Python to deliver invoicing, warehousing and order-related data.

Education

Fast University (2018-2022)

B.S. Computer Science, 3.6/4

Skills

Data engineering & storage

SQL,Python, PySpark, CDC (Change Data Capture), Kafka, Trino, dbt ,Airflow, Apache Iceberg, Hive Metastore, DuckDB, PostgreSQL, MySQL, Data Modeling, TimescaleDB, DocumentDB, Apache Ranger

Cloud & Infrastructure

AWS S3 ,RDS ,EC2 ,EKS ,EMR ,Glue ,Athena Redshift ,DMS ,VPC PrivateLink ,IAM ,Kubernetes ,Docker ,Helm ,Terraform ,ArgoCD ,Databricks, GCP (BigQuery),CI/CD

AI, observability & analytics

LangChain, RAG, CrewAI, Open WebUI, OpenCode, Grafana, Percona, Metabase, KEDA

Certifications

Introduction to Data Engineering

IBM 2025

Building Generative AI Applications

IBM 2025

Fundamentals of AI Agents Using RAG and LangChain

Coursera 2025

Awards

Dean List **2018-2022**

Fast University

Enlisted in Dean List **7 times** throughout degree

Achievements

2nd Position at AI Hackathon, Geeks of Kolachi

3rd Position in AI Hackathon, Bazaar Technologies